

SANGJUN (SAMUEL) Ko

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WORK EXPERIENCE

Kimberly-Clark — North America Family Care R&D

Statistician (contract assignment through Staffmark Group)

Neenah, Wisconsin

June 2025 – Present

- Serve as statistical lead for approximately 90% of 15+ completed and ongoing Test Method Validation (TMV) and measurement-system studies across multiple product lines and laboratory methods.
- Develop study designs, sample-size recommendations, randomized and blinded run orders, acceptance criteria, statistical analyses, and qualification recommendations for continuous and attribute-based methods.
- Coordinate closely with laboratory, test-method, product, and quality teams throughout study planning, execution, data review, and interpretation; analyze the resulting data and communicate findings and recommended next steps.
- Support TMV and measurement-system work involving basis weight, tensile, caliper, tear strength, attribute inspection, breathability, equipment equivalency, method transfer, and Gauge R&R.
- Lead statistical planning for product- and claim-supporting methods spanning R&D, quality, product release, and design verification, including collaboration with product, quality, laboratory, and legal stakeholders.
- Translate statistical results into recommendations for test-method design, equipment qualification, acceptance criteria, follow-up testing, and investigation of unexpected product-performance patterns; product and method owners retain final decision authority.
- Analyze an average of six requested mannequin leakage studies per month across two business units using ANOVA, mixed models, least-squares means, multiple comparisons, leak-location summaries, and sensitivity analyses.
- Interpret product performance and failure patterns for product release, prototype screening, material qualification, DOE, competitive benchmarking, and consumer-study support; explain results and recommended follow-up actions to researchers and project teams.
- Designed and deployed *leakageReport*, a Python/Flask reporting application that replaced a retiring legacy workflow and allows 50+ researchers and technicians to upload data and generate standardized mixed-model and pairwise-comparison reports in minutes rather than waiting several days.
- Build, deploy, and maintain six modules on an internal analytics platform, including leakage reporting, self-service predictive-model interfaces, an X-ray absorbency analysis port, a mannequin catalog and intake tool, and a usage-tracking dashboard; serve as a primary technical contact for the platform.
- Maintain and further train established product-performance models, coordinate laboratory testing to generate new data, personally design supporting experiments, analyze results, and operationalize actively used models through self-service interfaces that increase their visibility and accessibility across R&D.
- Design and analyze DOE-based studies for development of a Youth TMD mannequin screening method intended to replace portions of multi-month consumer home-use testing with a three-week screen, with projected multimillion-dollar savings communicated by project leadership.
- Lead development of a whole-product breathability data-reduction method and accompanying Python analysis pipeline, including recommendations for steady-state value selection and minimum run-time rules to improve consistency in future testing.
- Develop initial design options for a global interlaboratory comparison involving laboratories in the United States, China, and Vietnam, addressing relative bias, repeatability, reproducibility, sample coverage, and practical execution constraints.

STAT 107: Data Science Discovery, University of Illinois Urbana-Champaign

Undergraduate Course Assistant

Champaign, Illinois

August 2023 – May 2025

- Mentored 300+ students in foundational data-science concepts through interactive laboratory sessions and individualized technical support.
- Managed Git-based updates and created, edited, and proofread instructional content for a learning platform serving 1,000+ students.
- Collaborated with a staff of 100+ course assistants and two professors through weekly planning and coordination meetings.

STAT 443: Professional Statistics, University of Illinois Urbana-Champaign Champaign, Illinois
Undergraduate Course Assistant January 2025 – May 2025

- Graded assignments and provided feedback to 20+ students on statistical analysis, consulting, and communication.
- Evaluated 5+ student presentations for analytical quality and alignment with project expectations.
- Acted as a client in mock consulting projects, posing realistic technical questions and business constraints.

Applied Technologies for Learning in Arts & Sciences Champaign, Illinois
AI/ML Intern Team Leader January 2024 – December 2024

- Led a team of five interns and coordinated priorities, deliverables, and progress with three managers through weekly meetings.
- Developed and evaluated Python machine-learning models using datasets containing 100,000+ records.
- Applied regression, random forests, XGBoost, neural networks, BERT, and transformer-based methods to predictive and natural-language-processing projects.

HappySneeze Remote
Market Research Intern August 2024 – September 2024

- Created an interview guide and conducted 10+ consumer interviews concerning menopause and pelvic-floor health needs.
- Conducted market and consumer-trend research and synthesized findings into product and market recommendations for stakeholders.

KM Internacional S.A. Managua, Nicaragua
Intern May 2023 – August 2023

- Oversaw inventory management and streamlined administrative processes for a small textile company.
- Led three employees in improving record-keeping accuracy to 98% and reducing operational costs by 15%.

Korean Augmentation to the United States Army (KATUSA) Gyeonggi-do, South Korea
Soldier — Infantry and S-4 Logistics June 2021 – December 2022

- Completed an 18-month assignment in bilingual and cross-cultural U.S.–Korean military environments.
- Performed infantry responsibilities involving maintenance and operation of an M1 Abrams tank with the 2nd Squadron, 13th Cavalry Regiment.
- Supported S-4 logistics and inventory operations for the 176th Financial Management Support Unit, maintaining accurate and efficient accountability processes.
- Served as a Korean–English liaison and translator for 2,000+ U.S. and Korean soldiers during joint training exercises and assisted newly stationed personnel with local orientation.

Istmo Textiles S.A. Masaya, Nicaragua
Intern February 2021 – March 2021

- Studied corporate social responsibility, human rights, labor and environmental requirements, industrial safety, and textile import/export operations between Central and North America.
- Participated in community-support activities providing school supplies and food to schools and churches in Masaya.

PROJECTS AND AWARDS

Graduation Prediction Model September 2024 – December 2024

- Increased model AUC from 0.65 to 0.82 through feature engineering and outlier review across 50,000+ student records.
- Built a Streamlit interface using 12 key predictors to provide real-time graduation predictions for university administrators.

3rd Place, 2024 COVID Information Commons Student Paper Challenge September 2024

- Analyzed socioeconomic and mental-health effects of COVID-19 in Mexico using statistical and geospatial methods and proposed interventions for vulnerable groups.

Paraphrase Detection July 2024

- Achieved 90%+ accuracy on the MRPC dataset using BERT, RoBERTa, and SBERT and developed a Gradio interface for real-time evaluation.

International Student Forecasting April 2024

- Processed 100,000+ enrollment records, identified trends, and forecast international enrollment to support resource-allocation recommendations.

Titanic Machine-Learning Model

February 2024

- Developed an 80%-accuracy Python classification model and used feature importance to identify key survival factors.

EDUCATION

University of Illinois Urbana-Champaign

Champaign, Illinois

Bachelor of Science in Statistics, Mathematics Minor; GPA: 3.93/4.00

January 2023 – May 2025

- Relevant coursework: Applied Bayesian Analysis, Advanced Data Analysis, Statistical Modeling, Statistical Data Management, Statistical Consulting, Probability, Real Analysis, Linear Algebra, and Data Science Programming.

University of Michigan–Shanghai Jiao Tong University Joint Institute

Shanghai, China

Coursework toward a B.E. in Mechanical Engineering; transferred to UIUC September 2019 – December 2020

- First Class Scholarship.

SKILLS

Statistical methods:	DOE, TMV, Gauge R&R/MSA, mixed models, ANOVA, multiple comparisons, regression, attribute agreement, bias and precision analysis, equipment equivalency, method transfer
Programming:	Python, R, SQL, JavaScript, HTML/CSS, Excel/VBA
Libraries/frameworks:	Pandas, NumPy, SciPy, statsmodels, scikit-learn, TensorFlow, Flask, Streamlit, Gradio, Matplotlib, Seaborn, Tidiverse, ggplot2
Tools:	JMP, Git, Docker, SAS, Tableau, Microsoft Office
Languages:	Korean and Spanish (native); English (fluent); French and Chinese (basic)
Certificates:	Data Science DISCOVERY; Data Science Certificate; Coursera Business Analytics Certificate

LEADERSHIP, SERVICE, AND ACTIVITIES

Volunteer, Habitat for Humanity — Tornado Recovery

Menasha, Wisconsin

August 2026

- Cleared storm debris and checked in with households in the Menasha area that requested tornado-recovery assistance through 211 or Habitat for Humanity.

Volunteer, Appleton Public Library

Appleton, Wisconsin

August 2025 – Present

- Shelve and organize materials and support “Game On” programs through setup, participant guidance, and activity assistance.

Consultant, Illinois Business Consulting

Champaign, Illinois

September 2024 – December 2024

- Collaborated with an interdisciplinary team on market analysis, operational optimization, and growth-strategy recommendations for a client engagement.

Treasurer, Performance UIUC Touch

Champaign, Illinois

February 2023 – May 2024

- Planned and executed 10 fundraising initiatives, increasing available funds by 30% to support activities, costumes, and travel.
- Participated in weekly practices and public performances and helped expand the team’s social-media audience by 100+ followers.

President, Asian Culture Club

November 2018 – March 2019

- Led five students in promoting multicultural engagement at an international high school and organized weekly meetings and monthly activities.

Volunteer, Habitat for Humanity

Nicaragua

September 2016 – June 2017

- Supported home and sanitation-system construction and maintenance projects serving low-income communities, contributing to 15 homes.

Teacher, Nanum School

Masaya, Nicaragua

August 2016 – December 2016

- Taught English and mathematics to children ages 5–12 through a local community school organized by a Korean Presbyterian church.

ADDITIONAL HONORS

- Army Achievement Medal, United States Army, December 2022.
- First Class Scholarship, University of Michigan–Shanghai Jiao Tong University Joint Institute, 2019.
- Salutatorian and Mathematics Award, Notre Dame International School, 2019.